Applied Industrial Data Science

Week 3, Lecture 3: Data Quality and Validation

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WEEK 3

07:50. The file is on your desk. 08:00. The supervisor decides.

You cannot collect more data. What are you willing to say?

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Three verdicts, and one rule

01 **Ship**
Use it. The number is good enough for the decision on the table.

02 **Hold**
Do not use it yet. Something has to be answered first.

03 **Investigate**
Use part of it, and go and find one specific thing.

Every verdict is about one decision, not about the file.

Four extracts from a real export follow. Each one looks fine until you ask it the supervisor's question.

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A simple question, and a clean-looking table

Hour	Line	Units	Downtime	Energy kWh	Motor °C
09:00	Line_A	461	2.6	165.7	72.8
09:00	Line_B	356	19.6	143.9	74
09:00	Line_C	407		171.5	76.2
10:00	Line_A	437	7.9	167.4	74.9
10:00	Line_B	357	18	139.6	75.7

Which line had the most downtime this morning?

Ship, hold, or investigate?

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One blank cell. Two questions. Two answers.

The question	What the blank does	Verdict
Rank the lines by downtime	Line_C has no downtime reading. Skip the blank and it drops out of the ranking; read it as zero and it leads. Neither is flagged.	Hold
Rank the lines by units produced	Nothing. units_produced is present and correct in that row.	Ship

A missing value is not a property of the file.
It is a property of the question you asked it.

So "is this data good?" has no answer. "Good enough for what?" does.

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Two numbers that stand out

Hour	Line	Units	Downtime	Energy kWh	Motor °C
19:00	Line_A	396	3.4	152.2	73.1
19:00	Line_B	319	21.4	-12	74.6
19:00	Line_C	403	4.4	173.3	74.4
00:00	Line_A	394	9.4	157.7	75
00:00	Line_B	388	5.3	153.1	112

Same verdict for both rows, or different verdicts?

Ship, hold, or investigate, for each one?

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One is impossible. One is an outlier.

	energy_kwh = -12.0	motor_temp_c = 112.0
What it contradicts	The meaning of the column. A running line consumed energy.	Nothing. Only what you expected to see today.
Who can rule on it	You can, now.	A maintenance engineer, with the readings either side and the shift log.
If you delete it	You lose a broken reading.	You lose the only trace of the event the supervisor needs to hear about.
Verdict	Hold: reject the value, keep the row	Investigate: flag it, name who decides

That one reading moves Line_B's average motor temperature from 76.5 to 78.0, and puts it top of the list.

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Nothing is missing, and nothing is impossible

Hour	Line	Units	Downtime	Energy kWh	Motor °C
12:00	Line_B	396	12.6	163.2	75.7
12:00	Line_C	433	12.6	170.1	76.5
13:00	Line_A	450	6.7	161.8	74.5
12:00	Line_C	433	12.6	170.1	76.5

Every value here is plausible. Ship it?

Look at rows 2 and 4.

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Two kinds of duplicate

Exact duplicate

The same record twice.

Hour	Line	Units	Downtime
12:00	Line_C	433	12.6
12:00	Line_C	433	12.6

Every total for Line_C counts this hour twice.

Fix: delete one copy and log it.

Business-key duplicate

Same line and hour, different numbers.

Hour	Line	Units	Downtime
12:00	Line_C	433	12.6
12:00	Line_C	418	14.1

Two sources disagree. Which one is right?

Investigate: deleting one means picking a winner.

Both are caught by writing down what one row means.
One line, one hour.

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A different export, a month later

Hour	Line	Units	Downtime	Energy kWh	Motor °C
14:00	Line_A	429	6	169.9	71.7
14:00	Line_B	383	11.6	150.2	76.6
14:00	line_c	375	18.9	165.7	77
15:00	Line_A	446	7.8	173.5	74.3
15:00	Line_C	409	15.9	169	74.1

Two faults here. Neither one is a bad value.

Nothing is missing, impossible, or duplicated.

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Two faults you can only find by counting

What changed	What a person sees	What the computer sees
line_c instead of Line_C	The same line. Obviously.	A fourth line with one row, and Line_C with only one of its two hours.
Line_B has no 15:00 record	Nothing at all. No gap, no blank, no warning.	2 hours × 3 lines = 6 rows expected. There are 5.

No value is wrong. No value is missing.
Both faults are invisible unless you count and compare.

Schema and coverage. Both are waiting in your lab file.

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A timestamp that parses can still be wrong

Timestamp in the file	Line	What pandas reads
2026-01-12T14:00	Line_A	12 January, 14:00
12/01/2026 15:00	Line_A	1 December, 15:00
2026-01-12T16:00	Line_A	12 January, 16:00

Line_A's 15:00 hour leaves 12 January and turns up in December.
Nothing fails.

State the format when you read the file. With format="ISO8601", pandas stops on the odd row.

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All of this has a name

Dimension	The question it asks	Where it appeared
Completeness	Are the required values there?	The blank downtime on Line_C
Validity	Do the values contradict what the column means?	Energy of -12.0
Uniqueness	Does one business key identify one record?	Line_C at 12:00, twice
Schema	Are the expected fields and types present?	line_c instead of Line_C
Coverage	Are the required entities and periods all there?	Line_B with no 15:00 record
Timestamp	Do the times parse, sort, and cover each period once?	12/01 read as 1 December

Five of the six are answered by counting. Only validity needs someone with authority.

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The tempting fix

	Hour	Line	Energy kWh	Validator
Raw file	19:00	Line_B	-12.0	FAIL, CSV line 42
After a quick edit	19:00	Line_B	12.0	PASS

Green. Who would ever know?

Nothing in the file, the repository or the report now says a value was changed, what it was, or who decided.

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What makes evidence checkable by someone else

RAW SOURCE

Never edited, never overwritten. The only thing that can settle a dispute later.

PROCESSED LAYER

Corrections, exclusions and flags, each one linked back to the record it came from.

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SHA-256: a3f1... 8e2c      <- your file, and nobody else's
```

A file name is not provenance. A fingerprint, a source, and a date are.

Origin. Acquisition. Transformation. Use. Approval. Five lines in a README, written once.

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Three things people collapse into one

Observation	Interpretation	Action
What the file shows. No opinion in it at all.	What might explain it. Always more than one.	What has to happen, and who has to do it.
motor_temp_c is 112.0 at Line_B, 13 Jan 00:00. Every other reading in the file is 71 to 78.	A failing sensor. Or a real thermal event. Or a heavier product mix on that shift.	Maintenance engineer confirms the approved limit and checks the readings either side. Flagged, not deleted.

"112 degrees is a sensor fault" is an interpretation wearing an observation's clothes.

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A dataset decision is a sentence, not a label

This file **may** be used to compare units produced by each line across the day,
but **may not** be used to rank the lines on downtime, or judge the health of Line_B's motor,
on condition that the repeated Line_C 12:00 row is removed and logged,
and the decision is **revisited when** the missing downtime value is supplied and the engineer has ruled on the 112 °C reading.

Decision: Approve with conditions

01	02	03	04
Approve	**Approve with conditions**	**Reject pending replacement**	**Stop**
Meets the contract for the named use.	The normal answer: narrow use, visible issues.	Not this file, but a better one may exist.	The question or the risk rules the work out.

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A decision someone could act on

	Weak	Strong
May be used to	"for analysis" Analysis is not a use.	compare units produced by each line across the day
May not be used to	"for prediction" Names no claim.	rank the lines on downtime, or judge Line_B's motor
On condition that	"issues are noted" Noted where, and by whom?	the repeated Line_C 12:00 row is removed and logged
Revisited when	"later" Later is never.	the missing downtime arrives and the engineer rules on 112 °C

In the lab you write this for your own file.
It is 20% of the mark: specific enough to act on, and to be wrong.

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The row that passes all checks

On a scorecard
Every number is possible on its own.

Batter	Runs	Balls	4s	6s
Opener	85	0	9	3

Rule that catches it: runs > 0 needs balls > 0

In our file
Written to get past the validator.

Hour	Line	Units	Downtime
14:00	Line_A	400	60

Rule that catches it: downtime 60 → units 0

Every value is in range.
Together, they are impossible.

None of the twelve checks relates downtime to units.
Nobody writes that rule until a row like this gets past.

That is the last part of the lab. Bring me a better one.

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In the lab, the file is yours alone

IN THE LAB

Start from the six dimensions
Generate your own file from your username
Find the faults before the validator does
Write six checks until the tests pass
Then break your partner's validator

WEEK 4

Reproducible project architecture
The checks move into a repository
that runs them without you

Bring one rule you can defend, and one row you still cannot explain.

The second one is worth more marks than the first.

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